

Agentic Multimodal QA on Long Documents

From Parsing to answering

Hamza El Abdouni

M2 MVA (ENS Paris-Saclay) — BNP Paribas CIB - Analytics Consulting

September 2026

Outline

1. The agent and its tools
2. Results
3. A2: delegating orchestrator
4. Limitations & future work

Starting Point : Intelligent Document Processing

Internship subject : leverage multimodal LLMs for *information extraction* from visually rich documents.

First attempt : the mPLUG-DocOwl family (1 / 1.5 / 2) : a page image is sliced into crops encoded independently (a shape-adaptive “zoom”), then a compression module condenses each page to ≈ 324 tokens before feeding the LLM end-to-end.

Model	Acc	F1
mPLUG-DocOwl 1.5	6.9	6.3
mPLUG-DocOwl2-8B	13.4	8.9

MMLongBench-Doc

Why it breaks down : direct end-to-end injection, with *no selection/RAG* and *no iterative navigation*, loses attention on evidence *scattered* across a long document.

A Dead-End, Then a Method

- **Reframing** : instead of a new end-to-end model, build a *method* around the multimodal LLMs *already in the internal catalog*.
- **DocAgent** (Sun et al., EMNLP'25): an *agentic* use of multimodality that scales to *long documents*. $\Rightarrow$ focus shifts to long-document processing.
- **The catch** : DocAgent relies on a parsed *XML skeleton* to navigate (they use Adobe's PDF parser). $\Rightarrow$ this opens the parsing question.

Problem & Research Question

The reframed problem: long financial documents (100+ pages). Answering = find the right page/section (navigation) then read the text, and multimodal reading a table *or* a chart.

Research Question : Two Coupled Sub-Problems

- (a) **Parsing**: compress a long document into a *compact structured representation (XML Tree)* the agent can traverse.
- (b) **Agentic navigation**: an agent with access to that representation, deciding what to read and how to read it.

Contributions

- C1 Parsing & XML skeleton**: a modular CPU-only pipeline that compresses a long document into a compact, navigable structured representation.
- C2 A navigating agent that validates the parsing**: with `get_section` alone, benchmarked against `oracle_md` and the `fullctx` baseline.
- C3 Selective, multimodal access**: adding search (BM25) and `get_visual`, the agent reaches the multimodal.
- C4 An orchestrator that delegates reading**: it navigates the skeleton and hands off each section to a fresh sub-agent that reads it and reports back, same accuracy, fewer tokens.

Parsing : a Modular, CPU Pipeline

End-to-end VLM parsers (MinerU2.5, PaddleOCR-VL(1.5/1.6)) require a GPU. Choice : small specialized models with an **explicit layout** stage.

Layout detection → Cleaning → Regin based OCR → assembly (flow diagram)

- **Layout** : PP-DocLayoutV2 (RT-DETR), 25 labels, per-label confidence thresholds.
- **OCR** — Mistral OCR ; tables recovered as structured text.
- **Assembly** — geometric cleaning, reading-order fix, grouping into atoms & sections.

Parsing : a Modular, CPU Pipeline

End-to-end VLM parsers (MinerU2.5, PaddleOCR-VL(1.5/1.6)) require a GPU. Choice : small specialized models with an **explicit layout** stage.

Layout detection → Cleaning → Region-based OCR → Assembly

- **Layout** : PP-DocLayoutV2 (RT-DETR), 25 labels ; 300 raw boxes/page, filtered by per-label thresholds.
- **Cleaning** : resolve overlaps and nesting (e.g. cells inside a table) into a clean box set.
- **Region-based OCR** : each region is cropped, and the crops are stacked into one composite image, each tagged with a marker; Mistral OCR reads the whole image, and the markers re-attribute the text to each box.

Turning a Page Into Structure

After parsing, every page gives us four things per box: *where* it is (bbox), *what* it is (label), its *text*, and the *reading order*.

From this we rebuild the document's outline:

- **Sections** : a title box (doc_title / paragraph_title) starts a section; everything until the next title belongs to it.
- **Nesting** : which title is a chapter, which is a subsection? One LLM call reads all titles in order and assigns a level (1–4), so sections nest into a tree (idea borrowed from MinerU).
- **Repeated page headers** : a title reprinted at the top of every page would open a new section each time; we detect and drop these repeats.
- **Captions** : for tables and figures, we detect their caption by looking at the box right before or after: if it is a figure_title or footnote, it is attached as the caption, making the table or figure reachable by text search.

The Agent : Navigating the Skeleton

The agent sees the **skeleton only** (the map), and calls tools to fetch content on demand : instead of reading the whole document.

A loop: the model reads the skeleton, calls a tool, gets the result, decides the next step.

- `get_section` : open a section's full content
- `search` : lexical BM25 over the sections
- `get_visual` : crop a figure/table and send it to the agent.
- `submit_draft` : commit the final answer

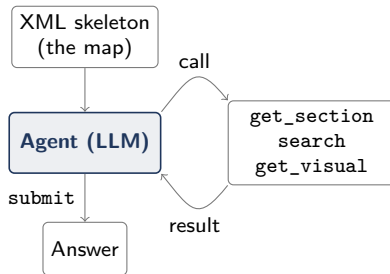


Diagram: agent loop
skeleton ↔ tools ↔ answer

Evaluation Protocol

Benchmark : FinRAGBench-V

- 96 financial PDFs, 467 questions
- 54 pages per document on average
- each question tied to *gold evidence pages*
- categories: tables, charts, text (single- & cross-page)

Scoring : LLM-as-judge

- answers are open-ended $\Rightarrow$ no exact match
- a judge model (**gemma-4**) grades answer vs. ground truth
- same judge, same prompt for *every* run

Reading the numbers

All runs use the same agent model (**mistral-medium-3.5**) and the same judge, so we only compare **deltas at constant setup**.

Diagnostic Method : Isolating Each Ceiling

To know *where* the agent loses, we compare it to controlled conditions where part of the problem is solved for it.

- **Baseline** `fullctx(text)` no navigation : the whole PDF is fed in *30-page chunks* (3-page overlap), then answers are merged. *How far does brute force go ?*
- **Oracles** : we hand the LLM the gold pages. Retrieval is solved, so we measure reading alone. Three modalities isolate what reading needs :
 - ▶ `oracle_md` : gold pages as text only
 - ▶ `oracle_image` : gold pages as image only
 - ▶ `oracle_both` : text + image

These give the ceilings : how good could the agent be if navigation were perfect ? And does the answer live in the text, the image, or both ?

Validating the Skeleton : `get_section` Alone

First agent run : `get_section` only, no `search`, no `get_visual`. Can it navigate the skeleton and answer ?

Run	Overall	tok/q	TAB-Num	TAB-Cmp	TXT-Inf	CHA-Info	CHA-Time	CHA-Num	TAB-MP	TXT-MP
fullctx (baseline)	41.3	124K	39.4	55.7	51.9	41.3	27.9	5.4	37.9	50.0
<code>get_section</code>	46.7	108K	52.3	62.0	59.5	31.7	18.6	2.7	58.6	67.9
oracle_md	55.0	1.6K	69.7	68.4	73.4	34.9	18.6	2.7	58.6	75.0

- **Navigation beats brute force** : on text & tables, `get_section` clearly tops `fullctx` (TAB-MP 37.9→58.6, TXT-MP 50→67.9): the skeleton is usable.
- **A navigation gap remains** : 8.3 pp below `oracle_md`: with no `search`, the agent doesn't always find the right section. ⇒ add `search`.
- **Charts stay low** : a chart is a *multimodal* query: accuracy isn't zero only because a few answers can be found in the page's text or on other pages. ⇒ add `get_visual`.

Adding search : Closing the Navigation Gap



search (BM25) lets the agent *find* sections, not just open known ones. Compared to the three oracles (gold pages given):

Run	Overall	tok/q	TAB-Num	TAB-Cmp	TXT-Inf	CHA-Info	CHA-Time	CHA-Num	TAB-MP	TXT-MP
get_section	46.7	108K	52.3	62.0	59.5	31.7	18.6	2.7	58.6	67.9
+ search	52.0	106K	64.2	68.4	69.6	33.3	9.3	2.7	58.6	75.0
oracle_md	55.0	—	69.7	68.4	73.4	34.9	18.6	2.7	58.6	75.0
oracle_image	60.0	—	41.3	64.6	74.7	77.8	48.8	45.9	62.1	71.4
oracle_both	66.4	—	68.8	73.4	70.9	79.4	41.9	45.9	55.2	71.4

- **Search closes most of the navigation gap** — 46.7 → 52.0, now near oracle_md (55.0) at similar token cost.
- **But text has a ceiling** — on charts, both the agent *and* oracle_md stay low ; only the *image* oracles jump (CHA-Info 34.9 → 79.4). ⇒ the answer is in the pixels.

Adding `get_visual`: Reading the Pixels



`get_visual` crops a figure/table and OCRs it — the agent now reads the image *when it chooses to*.

Run	Overall	tok/q	TAB-Num	TAB-Cmp	TXT-Inf	CHA-Info	CHA-Time	CHA-Num	TAB-MP	TXT-MP
full agent (k=5)	65.7	102K	67.9	68.4	67.1	66.7	53.5	51.4	69.0	78.6
oracle_image	60.0	—	41.3	64.6	74.7	77.8	48.8	45.9	62.1	71.4
oracle_both	66.4	—	68.8	73.4	70.9	79.4	41.9	45.9	55.2	71.4

- **The full agent reaches the multimodal ceiling** : 65.7 vs oracle_both 66.4 (−0.7 pp), *while navigating itself*.
- **It even beats oracle_both on charts** : CHA-Time 53.5 vs 41.9, CHA-Num 51.4 vs 45.9: a targeted crop reads better than dumping the whole page. (small n)
- **And keeps text where text wins** : no collapse on dense tables (TAB-Num 67.9), unlike oracle_image (41.3). ⇒ conditional routing beats brute-force fusion.

A Navigation Example

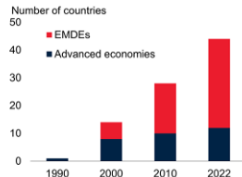
Q: *How has the number of countries with inflation targeting changed from 1990 to 2022?*

category: Chart-Time

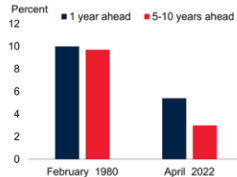
FIGURE SF1.7 Developments in the 1970s and 2020s: Structural differences

Inflation expectations are better anchored than in the 1970s. Inflation targeting and lower energy intensity may help contain the transmission of surging global energy prices into core inflation. Wage-setting arrangements are less centralized now than in the 1970s.

A. Number of countries with inflation targeting



B. U.S. inflation expectations



C. Labor market flexibility



D. Global energy intensity



A Navigation Example

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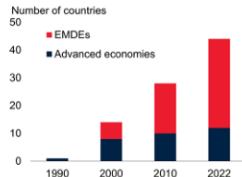
1. search("inflation targeting countries 1990 2022")

→ top hit s93

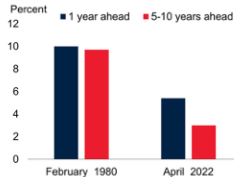
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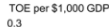
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category: Chart-Time

1. `search("inflation targeting countries 1990 2022")`
→ top hit s93
2. `get_section(s93)`
→ 4 charts inside, picks the right one

```
<section id="s93" level="4" pages="84-85" title="FIGURE SF1.7 Developments in the 1970s and 2020s: Structural differences" n_content="7" n_figures="4">  
  <preview>Inflation expectations are better anchored than in the 1970s. Inflation targeting and lower energy intensity may help...  
</preview>
```

A Navigation Example

Q: *How has the number of countries with inflation targeting changed from 1990 to 2022?*

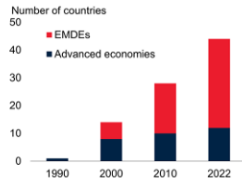
category: Chart-Time

1. `search("inflation targeting countries 1990 2022")`
→ top hit s93
2. `get_section(s93)`
→ 4 charts inside, picks the right one
3. `get_visual(p84_e9)`
→ crops + reads the chart

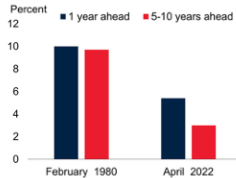
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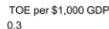
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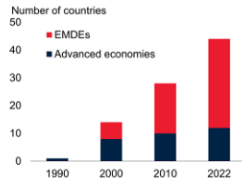
1. `search("inflation targeting countries 1990 2022")`
→ top hit s93
2. `get_section(s93)`
→ 4 charts inside, picks the right one
3. `get_visual(p84_e9)`
→ crops + reads the chart
4. `submit_draft ✓`

4 steps, 7.9s — navigation + reading, no full-document read.

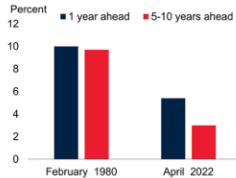
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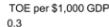
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Beyond FinRAGBench : Public Calibration (MMLongBench-Doc)

A second, general-domain benchmark — to place the agent against published systems.

System	Backbone	Acc	F1
Ours (A1)	mistral-3.5 (generalist)	48.06	44.86
DocAgent (Sun)	Qwen3-VL-30B	48.3	45.8
GPT-4o (non-agentic)	GPT-4o	—	44.9
MARDoc	—	57.1	54.6
AgenticOCR	Qwen3-VL-8B	66.4	—
DocLens	Gemini-2.5-Pro	67.6	—

- **On par with the agentic SOTA that uses a dedicated VLM** (DocAgent/Qwen3-VL-30B), with an *imposed generalist backbone*.
- **Above the best non-agentic baseline** (GPT-4o): the agentic structure pays off.
- Frontier-VLM systems (DocLens, AgenticOCR) lead — a *backbone* gap, not an architectural one.

30% stratified sample (323/1082) ; non-strict comparison (sample, backbone, scoring differ).

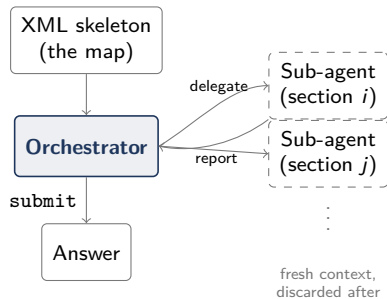
A2: an Orchestrator That Delegates Reading

In A1, one agent both *navigates* and *reads* — its context keeps growing.

A2 splits the two roles :

- **Orchestrator** — sees the *skeleton only*, decides what to read.
- **Sub-agent** — spawned *fresh* for one section, reads it, reports back, then is discarded.
- Only a short **report** returns to the orchestrator — not the raw content.

⇒ the orchestrator's context stays small: same accuracy, fewer tokens.



A2: Same Accuracy, Fewer Tokens

A2 (orchestrator + sub-agent, 6/6 rounds) vs. A1, same model and judge throughout.

	FinRAGBench-V		MMLongBench-Doc	
	Acc	tok/q	Acc	tok/q
A1 (full agent)	65.7	102K	48.06	58.4K
A2	67.5	84K	49.5	53.6K
Δ	+1.8	-18%	+1.5	-8%

- **Fewer tokens on both benchmarks** — the orchestrator's context no longer grows with every section read.
- **Accuracy holds** — +1.8 / +1.5 pp: the robust gain is **cost**, not accuracy.
- On FinRAGBench, A2 also **reaches** `oracle_both` (67.5 vs 66.4) .

Limitations

- **A reading floor remains** — even `oracle_both` tops out at 66.4: with gold pages given, a third of questions are still missed. Part of the gap is *reading*, not navigation.
- **Cross-page reasoning is weak** — the pipeline has no dedicated mechanism to combine evidence spread across pages.
- **Judge sensitivity** — absolute scores shift with the judge model; only *deltas at constant judge* are reliable here.
- **Inter-run variance ≈ 2.3 pp** — the LLM and judge are non-deterministic; small gaps are inconclusive, and variance was not fully quantified.
- **Intra-document setup** — one relevant document per query; cross-document retrieval is not evaluated.

Future Work

- **Cross-page handling** : let the agent gather and combine evidence across several pages.
- **Verification metrics** : flag hallucinated numbers or pages.
- **A third tier : a reviewer agent** : after the orchestrator drafts an answer, a reviewer checks it against the cited evidence and can send it back for another read (orchestrator → sub-agent → reviewer).

Conclusion

- **Parsing** — a modular, CPU-only pipeline compresses a long document into a compact, navigable **XML skeleton**.
- **A navigating agent** — validated against oracles and the `fullctx` baseline: with `search + get_visual`, it **reaches the multimodal ceiling** (65.7 vs 66.4) and, by routing modality conditionally, **beats both mono-modal oracles**.
- **A diagnostic method** — oracles isolate navigation, textual and visual reading as separate ceilings, turning “where does it fail?” into measurable quantities.
- **A2** — a delegating orchestrator holds accuracy at **markedly lower token cost**.

Thank you — Questions ?